

Department of Information Technology
Central University of Kashmir

B.Tech. CSE

Semester IV					
S.No.	Course Code	Course Title	Credits	CIA	ESE
1	BT 401	Mathematics IV (Probability and Statistics)	3	50	50
2	BTCS 408	Design and Analysis of Algorithms	4	50	50
3	BTCS 403	Discrete Structures	4	50	50
4	BTCS 404	Digital Electronics & Logic Design	3	50	50
5	BTCS 406 L	Digital Electronics & Logic Design Lab	2	50	50
6	BTCS 407	Economics for Engineers	2	50	50
7	BTCS409	Object-oriented Programming	4	50	50

8	BTCS 410	OOP Lab	2	50	50
		Total	24		

Central University of Kashmir

Course Title	Mathematics (Probability and Statistics)	Course Code	BT 401	
Degree	Bachelor of Technology	Branch	Computer Science & Engg.	
Course Type	Theory	Department	Information Technology	
Credits	04	L	T	P
Semester	4th	4	0	0

Course Outline / Content

Unit	Topics	Weeks
1.	Probability and Statistics: Measures of Central Tendency: Mean, Median, Mode, Dispersion: Range, Standard Deviation; Probability: Definition, Motivation, Probability Models, Probability Rules, Sample Space having equally likely outcomes, Conditional Probability; Baye's Theorem; Independent Events.	4
2.	Random Variable; Discrete Random Variable, Distribution Function, Probability Mass Function, Poisson and Binomial Distributions, Continuous Random Variables, Probability Density Function, Uniform and Normal Distributions.	4
3.	Correlation, coefficient of correlation, Linear Regression, regression coefficient, Method of Least Squares, Introduction to Sampling and Sampling Distribution, Standard Error, sampling distribution of the Mean (known) Sampling Distribution of the mean (unknown) – The sampling distribution of the variance.	4
4.	Hypothesis: Tests of Hypotheses, Null Hypotheses and Alternate Hypothesis, Critical Region, Type I and Type II Errors,, Significance tests, Level of Significance, Z-test and, χ^2 —Test of Goodness of Fit.	3

References

1.	Miller and Fread, “Probability and statistics for engineers – Richard A Johnson” Pearson Education Asia / PHI.
2.	Ross, S.M, “Probabilty and Statics for Engineers and Scientists”, 4th Edition, Elsevier.
3.	S.C.Gupta & V.K.Kapoor, “Fundamentals of Mathematical Statistics” Sultan & Sons, Eastern Economy Edition.
4.	Kishor S. Trivedi , “Probability And Statistics With Reliability, Queuing With Reliability, Queuing And Computer Science Application”, PHI.

Central University of Kashmir

Course Title	Design and Analysis of Algorithms	Course Code	BTCS 408	
Degree	Bachelor of Technology	Branch	Computer Science & Engg.	
Course Type	Theory	Department	Information Technology	
Credits	04	L	T	P
Semester	4th	4	0	0

Course Objectives

1. To analyse the asymptotic performance of algorithms.
2. To write rigorous correctness proofs for algorithms.
3. To demonstrate a familiarity with major algorithms and data structures
4. To apply important algorithmic design paradigms and methods of analysis.
5. To synthesize efficient algorithms in common engineering design situations.

Learning Outcomes

1. For a given algorithms analyse worst-case running times of algorithms based on asymptotic analysis and justify the correctness of algorithms using Divide and Conquer.
2. Apply the greedy paradigm and explain when an algorithmic design situation calls for it. For a given problem develop the greedy algorithms.
3. Apply the divide-and-conquer paradigm and explain when an algorithmic design situation calls for it. Synthesize divide-and-conquer algorithms. Derive and solve recurrence relation.
4. Apply the dynamic-programming paradigm and explain when an algorithmic design situation calls for it. For a given problems of dynamic-programming and develop the dynamic programming algorithms, and analyse it to determine its computational complexity.

5. For a given model engineering problem model it using graph and write the corresponding algorithm to solve the problems.

6. Compute the approximation factor of an approximation algorithm.

Course Synopsis

1. Introduction to algorithms: the basic concepts of algorithms, including what they are, how they are designed, and how they are analysed.

2. Asymptotic analysis measure the running time of algorithms using asymptotic analysis, including big-O, big-omega, and big-theta notation.

3. Sorting algorithms: different sorting algorithms, such as bubble sort, insertion sort, selection sort, quicksort, merge sort, and heap sort. They will understand the principles behind each algorithm and their time complexity.

4. Divide and conquer algorithms: divide and conquer algorithms, which involve dividing a problem into smaller sub-problems, solving each sub-problem.

5. Greedy algorithms: Students will learn about greedy algorithms, which make locally optimal choices at each step to achieve a global optimum.

6. Dynamic programming: dynamic programming, which involves breaking a problem into smaller sub-problems and storing the results of each sub-problem to avoid redundant computations

Course Outline / Content

Unit	Topics	Weeks
1.	Introduction: Algorithm Design paradigms- motivation, concept of algorithmic efficiency, run time analysis of algorithms, Asymptomatic Notations. Divide & Conquer: Structure of divide and conquer algorithms: 4 examples, Binary search, Quick sort, analysis of divide and conquer run time recurrence relations.	4

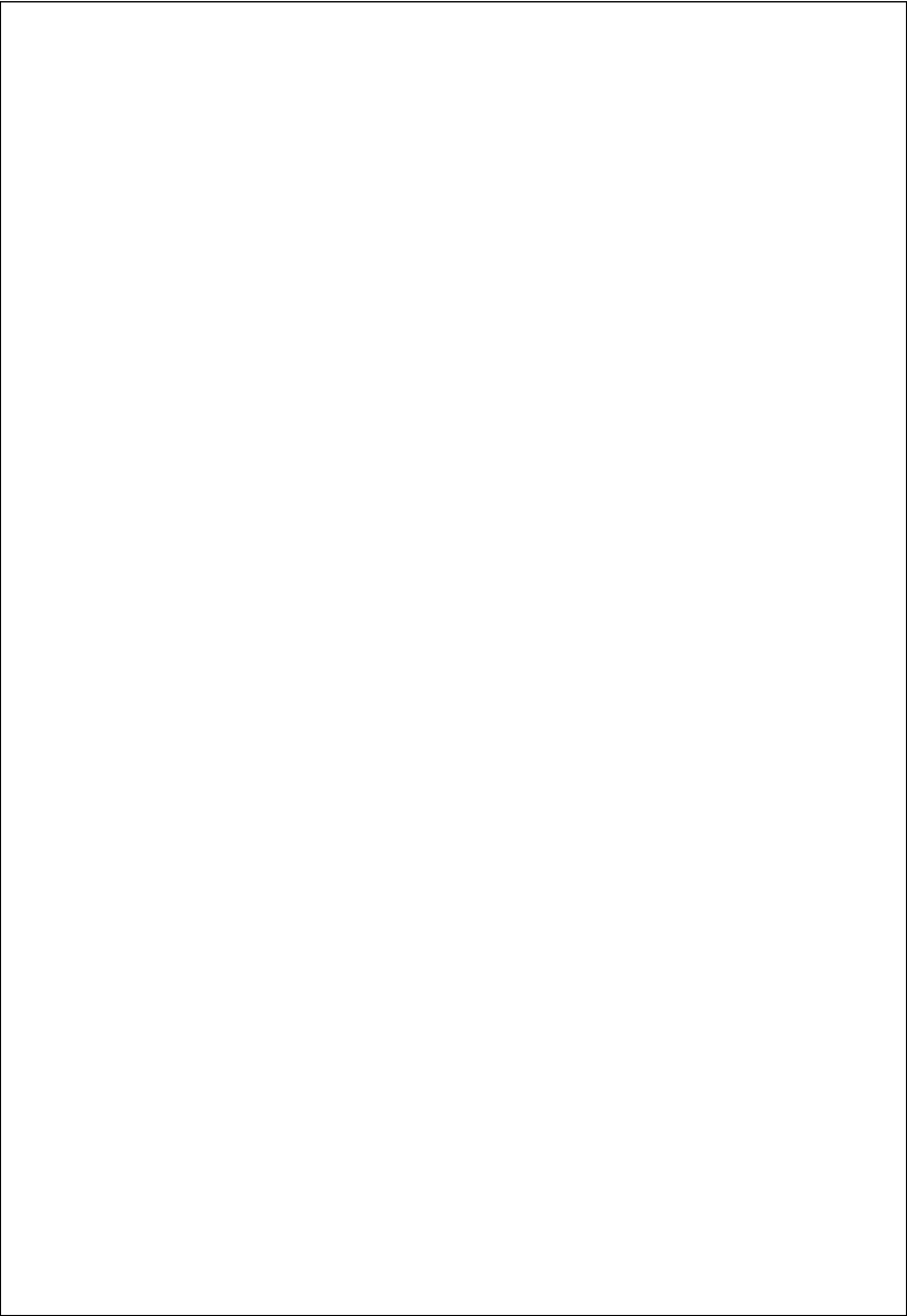
2.	Greedy method: Overview of the greedy paradigm, examples of exact optimization solution (minimum cost spanning tree), approximate solution (Knapsack problem), single source shortest paths .Dynamic Programming: Overview, difference between dynamic programming and divide and conquer, applications: shortest path in graph, matrix multiplication, travelling salesman problem, longest common sequence.	3
3.	Graph searching and traversal: Overview, traversal methods, depth first and breadth first search. Back Tracking: Overview, 8-queen problem and Knapsack problem.	4
4.	Branch & Bound: LC searching, bounding, FIFO branch and bound, Applications: 0/1 Knapsack problem, Travelling salesman problem. Computational complexity: Complexity measures, Polynomial vs Non Polynomial time complexity; NP hard and NP complete classes, examples.	4

Text Books

1.	Ellis Horowitz and Sartaz Sahani, “Computer Algorithms”, Galgotia Publications.
2.	Narasimha Karumanchi, “Data Structures and Algorithms Made Easy: Data Structures and Algorithmic Puzzles” Careermonk Publications; 5th ed. Edition

References

1.	T. H. Cormen, C. E. Leiserson, R. L. Rivest, Clifford Stein, “Introduction to Algorithms”, 2nd Ed., PHI.
2.	V. Aho, J. E. Hopcroft, J. D. Ullman, “The Design and Analysis of Computer Algorithms”, Addison Wesley.
3.	D. E. Knuth, “The Art of Computer Programming”, 2nd Ed., Addison Wesley.
4.	Algorithm Design, 1ST Edition, Jon Kleinberg and ÉvaTardos, Pearson.



Central University of Kashmir

Course Title	Discrete Structures	Course Code	BTCS 403	
Degree	Bachelor of Technology	Branch	Computer Science & Engg.	
Course Type	Theory	Department	Information Technology	
Credits	04	L	T	P
Semester	4th	4	0	0

Course Objectives

1. To understand the Principles of Predicate and Boolean Logic, Graph theory, Lattices, Group theory and Set theory for analysis of various computer science applications.
2. To understand the applicability of basic discrete structures in various fields of Computer science.

Learning Outcomes

1. Apply concept of Predicate Calculus in computer science like design of computing machines, Artificial intelligence, definition of Data structures for programming languages etc.
2. Apply the knowledge of Boolean algebra in computer science for its wide applicability in switching theory, building basic electronic circuits and design of digital computers.
3. Apply Graph algorithms to real-world problems: to identify and model real-world problems as graph problems.
4. Apply Graph theory to data analysis and visualization: to apply graph theory to analyze complex data structures, including social networks, biological networks, and transportation networks.
5. Apply set theory and relations to solve problems in computer science, such as database management, data structures, and network analysis. This includes the

ability to identify and represent data structures and relationships using sets and relations.

6. Apply abstract concepts of Group theory in modelling and solving non-trivial problems in different field of cryptography and coding theory.

Course Synopsis

1. Construct mathematical arguments using logical connectives and quantifiers.
2. Verify the correctness of an argument using laws of propositional and predicate logic and truth tables.
3. Standard forms of representation of Boolean function.
4. Minimization of Boolean expressions for construction of efficient circuits.
5. Use graphs and trees as tools to visualize and simplify situations.
6. The course in Discrete Mathematics will provide students with a strong foundation in mathematical concepts that are essential in computer science and information technology fields.
7. The students will develop their analytical and problem-solving skills, which are important for designing and analysing algorithms, cryptography, and artificial intelligence..

Course Outline / Content

Unit	Topics	Weeks
1.	Unit 1: Propositional Calculus Statements, Basic operations, Truth value of compound statements, Algebra of Propositions, Tautologies and contradiction, Conditional and Bi-conditional statements, logical implications, logical equivalence, predicates, Universal and existential quantifiers. Logic gates, Boolean Algebra, Postulates of Boolean Algebra; Theorems of Boolean Algebra, Sum of products and product of sums Simplification, NAND and NOR implementation	4

2.	Graph Theory Fundamental concepts (basic definitions, operations, properties); Trees (properties, distances and centroids, spanning trees, enumeration); Matchings (bipartite graphs, general graphs, weighted matching); Connectivity (vertex and edge connectivity, cuts, blocks, k connected graphs, network flows); Traversability (Eulerian tours, Hamiltonian cycles); Coloring (vertex and edge coloring, chromatic number); Planarity (duality, Euler's formula, characterization, 4-color theorem).	4
3.	Sets: Introduction, Set notations and description, sub-sets, Basis set operations, Venn diagrams, Combination of sets, Finite and infinite sets , Uncountable infinite sets, Mathematical induction, Principle of inclusion and exclusion. Relations: Definition, Properties of binary relations, Equivalence relations and partitions, Partial ordering relations. Hasse Diagrams.	3
4.	Group Theory Groups, semi group, infinite group, Finite group, order of a group, Abelian group, subgroup, Necessary and sufficient condition for a subset to be a subgroup of a group	4

Text Books

1.	Kenneth H. Rosen : Discrete Mathematics and its applications,5th Ed. Tata McGraw Hill (2003
2.	J L Mott, A Kandel, T P Baker, Discrete Mathematics for Computer Scientists & Mathematicians, Prentice-Hall of India.

References

1.	C. L. Liu, Elements of Discrete Mathematics, McGraw Hill International Editions. 5. K. D. Joshi, Foundations of Discrete Mathematics, Wiley Eastern Ltd.
2.	K.R Parthasarty : basic Graph Theory, Tata Mc-Graw Hill

- | | |
|----|--|
| 3. | B. Kolman and R.C. Busby: Discrete mathematical structures for computer science Prantice Hall, New-Delhi |
|----|--|

Central University of Kashmir

Course Title	Digital Electronics and Logic Design	Course Code	BTCS 404	
Degree	Bachelor of Technology	Branch	Computer Science & Engg.	
Course Type	Theory	Department	Information Technology	
Credits	04	L	T	P
Semester	4th	4	0	0

Course Objectives

1. To understand the basics of Digital Electronics and different number systems and conversion between them.
2. To Design and construction of the basic and universal logic gates.
3. To Stud the Boolean algebra and simplification of Boolean expression using different methods.
4. To Study and construction of sequential logic circuits, understanding various design of flip flops.
5. To Study the programmable logic devices, shift registers, counters and various memory devices..

Learning Outcomes

1. Apply Optimization for logical equations using reduction techniques
2. Design different types of code convertors
3. Construct Combinational and Sequential circuits
4. Validate the internal structure of combinational circuits
5. Develop applications of sequential circuits

Course Synopsis

1. The Course includes the Binary Systems, Boolean algebra & Logic Gates, Simplification of Boolean Functions, Combinational Logic, Sequential Logic, Registers, Counters, Data Converters.

2. The course will provide students with a strong foundation in designing and analyzing digital circuits and systems, which is essential in computer science and information technology fields.

3. The students will also develop their analytical and problem-solving skills, which are important for designing and optimizing digital circuits and systems.

Course Outline / Content

Unit	Topics	Weeks
1.	Review of number systems, BCD, Excess-3, Gray and Alphanumeric codes. Review of Boolean algebra, De-Morgan's Theorems, Standard Forms of Boolean Expressions, Don't care conditions, Minimization Techniques: K-MAPS, Q-M (Tabulation) method.	4
2.	Combinational Logic Circuits: Problem formulation and design of Basic Combinational Logic Circuits, Combinational Logic Using Universal Gates. Basic Adders, ALU, Parity-Checkers and Generators, Comparators, Decoders, Encoders, Code Converters, Multiplexer (Data Selector), Demultiplexers.	4
3.	Sequential Circuits: Latches, Flip-flops (SR, JK, T, D, Master/Slave FF,) Edge-Triggered Flip-Flops, Flip-Flop Operating Characteristics, Basic Flip-Flop Applications, Asynchronous Counter Operation, Synchronous Counter Operation, Up/Down Synchronous Counters.	4
4.	Shift registers & Memories, Shift Register Functions, Serial In - Serial Out Shift Registers, Serial In -Parallel Out Shift Registers, Parallel In - Serial Out Shift Registers, Parallel In - Parallel Out Shift Registers, Bidirectional Shift Registers.	4

	Basics of Semiconductor Memories, Random-Access Memories (RAM), Read Only Memories (ROMs).	
Text Books		
1.	Morris Mano, Digital Logic Design, TMH	
2.	Kumar Anand, Digital Logic Design, PHI.	
References		
1.	Thomas L. F., Digital Fundamentals, Prentice Hall, Inc, 4th Edition 1997	
2.	Tocci R. J. & Widner, Digital Systems: Principles and Applications, PHI.	
3.	Gothman, Fundamentals of Digital Electronics, PHI	

Central University of Kashmir

Course Title	Digital Electronics and Logic Design Lab	Course Code	BTCS 404 L	
Degree	Bachelor of Technology	Branch	Computer Science & Engg.	
Course Type	Theory	Department	Information Technology	
Credits	02	L	T	P
Semester	4th	0	0	2

Course Objectives

1. To understand the basic of Digital Electronic concepts required in analysis and design of digital electronic circuits and systems.
2. To understand the implementation of Boolean expression via different components.
3. To understand Construction and operation of various digital circuits such as Adder, Subtractor, Multiplexer, Demultiplexer, Decoder, Encoder, Flip-flops, Counters, Registers and memory devices.
4. To evolve the capability in student so that he/she will be able to simplify, Analyse and design the Various Digital Electronic Circuits.

Learning Outcomes

1. Apply knowledge of Boolean algebra and other minimization techniques for digital circuit design.
2. Identify, formulate and solve a problem based on combinational and sequential circuits.
3. Select the appropriate hardware and software tools for combinational and sequential circuit design.
4. Design and analyze combinational and sequential circuits for various practical problems using basic gates and flip flops.

Course Synopsis

1. The Digital Logic Design Lab will provide students with practical experience in designing, implementing, and testing digital circuits and systems.

2. The lab exercises will help students to develop their problem-solving and critical thinking skills, which are important for designing and optimizing digital circuits and systems.

Course Outline / Content

Lab		Weeks
	• Implementation of the logic of a. OR gate b. AND gate c. Not d. NAND gate e. NOR gate f. Ex-OR g. Ex-Nor • Realization of NAND and NOR as universal Gates. • To simplify the given expression and to realize it using Basic gates and Universal gates. • To realize a. Half Adder and Full Adder b. Half Subtractor and Full Subtractor • To design and set up a Multiplexer (MUX) , Demultiplexer (DE-MUX)	In acordance with theory of BTCS 404

	• To realize the logic of gates and the given Boolean expression via Mux. • Truth Table verification of 1) RS Flip Flop 2) T type Flip Flop. 3) D type Flip Flop. 4) JK Flip Flop. • Implementation of Synchronous, asynchronous counters and Modulo-Counters.	
Text Books		
1.	Morris Mano, Digital Logic Design, TMH	
2.	Kumar Anand, Digital Logic Design, PHI.	
References		
1.	Thomas L. F., Digital Fundamentals, Prentice Hall, Inc, 4th Edition 1997	
2.	Tocci R. J. &Widner, Digital Systems: Principles and Applications, PHI.	
3.	Gothman, Fundamentals of Digital Electronics, PHI	
	Simulation Tools	
1	PSpices and NGSpice	
	Xcircuit	

Central University of Kashmir

Course Title	Economics for Engineers	Course Code	BTCS 407	
Degree	Bachelor of Technology	Branch	Computer Science & Engg.	
Course Type	Theory	Department	Information Technology	
Credits	02	L	T	P
Semester	4th	02	0	0

Course Objectives

1. To understand major principles of economic analysis for decision making among alternative courses of action in engineering.
2. To understand the cost estimation and alternative analysis techniques for engineering applications.
3. To demonstrate knowledge of cost estimation techniques and probabilistic risk analysis.

Learning Outcomes

1. Apply the basic concepts and terminology used in engineering economics.
2. Apply the concepts of cash flows, time value of money in evaluation of investments and projects in real life.
3. Compare and evaluate alternatives based on present, annual, rate of return, and benefit over cost analyses.
4. Identify and analyse the impact of depreciation, taxation and other economic factors on feasibility of real life projects.
5. Recognize the economic impact of engineering solutions and Conduct sensitivity analysis on key compounding parameters, so as make financially prudent decisions in everyday life.

Course Synopsis

1. This course will introduce various concepts and methods of economic analysis in engineering, including the time value of money and its effect on economic decisions.

2. Course also discusses economic equivalence, economic measures of worth, cash flow analysis, equipment depreciation, effects of inflation, decision making under uncertainty, capital budgeting, replacement decisions, and benefit-cost analysis.

Course Outline / Content

Unit	Topics	Weeks
1.	Utility analysis –Concept of demand and law of demand; Cardinal Utility, Law of diminishing marginal utility; Ordinal utility— assumptions, Indifference curves and their properties, Marginal rate of substitution, Budget line, Consumer's equilibrium using Indifference curves; Elasticity of Demand— Price, income and cross elasticity.	4
2.	National income-concept and related aggregates, measurement of national income- income, expenditure and value added method; Circular flow of income and expenditure-two, three and four sector model; Consumption function and its determinants..	03
3.	Meaning and scope of public finance; Concept of budget and its types, measures of budget deficit- fiscal deficit, revenue deficit, and primary deficit. Taxation- types of taxes, Impact and incidence of taxation. Meaning and objectives of fiscal policy, Fiscal responsibility and budget management Act; Deficit Financing– meaning, objectives and causes, Effects of deficit financing.	03

References

1.	A. Koutsoyiannis, Modern Microeconomics, Second Edition, Macmillan Press, London.
2.	D.N Dwivedi, Microeconomics – Theory and Applications, Pearson Publications.
3.	P.A Samuelson and W. Nordhaus, Economics, 19th Edition, Tata McGraw Hill Publication.
4	Richard T Froyen, Macroeconomics, 10th Edition, Pearson Education. Delhi.
5	N. Gregory Mankiw, Macroeconomics, 5th Edition, Macmillan.
6	D.N Dwivedi, Macroeconomics – Theory and Policies, Pearson Publications.
7	H. L Bhatia, Public Finance, 29th Edition, Vikas Publication.
8	R.R Barthwal, Industrial Economics- An Introductory Textbook, New Age Publication.

Central University of Kashmir

Course Title	Object-oriented Programming	Course Code	BTCS 409	
Degree	Bachelor of Technology	Branch	Computer Science & Engg.	
Course Type	Theory	Department	Information Technology	
Credits	04	L	T	P
Semester	4th	4	0	0

Course Objectives

- Programming in the Java programming language.
- The knowledge of object-oriented paradigm in the Java programming language.
- The use of Java in a variety of technologies and on different platforms.

Learning Outcomes

- Students will be able to program using Java programming language.
- Knowledge of the structure and model of the Java programming language.
- Design, analyse and evaluate computer programs using the Java programming language.
- To use the Java for various programming technologies.
- Develop software in the Java.

Course Synopsis

- The course is divided into four units covering appropriate programming concepts related to Java.
- Object-oriented concepts-constructors, encapsulation, inheritance and polymorphism.
- Java code utilities in applets, packages and classes.

- Exception handling in Java programs.

Course Outline / Content

Unit	Topics	Weeks
1.	Introduction to Java Language: Creation of Java. How Java changed the Internet. Features of Java Language. Evolution of Java. Comparison with other languages like C++.Java Virtual Machine (JVM) and Byte-code. Java Language Overview: Lexical issues – Whitespace, Identifiers, Keywords, Literals, Separators, and Comments. Installing JDK.PATH variable. Java program – Structure, Compilation and Execution. Java Class libraries (System Class).main() method. Data types, Variables and Arrays: Primitive Data-types and Typed-Literals. Variables – Declaration, Initialization, Scope and Lifetime. Arrays – Single and Multidimensional. Type Conversion and Expression Promotion. Operators, Expressions and Control statements: Arithmetic, Bitwise, Relational, Logical, Assignment. Precedence and Associativity. Selection, Iteration and Jump Statements.	4

2.	Class Fundamentals: Class Structure (Variable and Method declaration).Modifiers (Access Modifiers and Other Modifiers).Components of Class, Variable and Method declaration. Constructor and finalize(). Garbage Collection. Passing parameters to methods. Variable hiding. Method overloading. Constructor overloading and chaining. Use of this keyword. Code blocks - Static and non-static. Inheritance: Mechanism. Role of Access Modifiers. Method Overriding and Shadowing. Use of super keyword. Polymorphism - Early and Late binding. Abstract Class and Interface. Components of Interface declaration. Implementing Interfaces. Exception Handling: Mechanism - Exception-Object, Throwing an Exception, and Exception Handler. Catch or Specify policy. Types of Exception - Checked vs Unchecked, Built-in vs User-defined. Catching an Exception - try-catchfinally. Specifying an Exception - throws. Manually throwing an Exception - throw. Custom Exceptions. Chained Exceptions..	5
3.	Packages: Creating and Importing Packages. CLASSPATH variable. static import. Strings: Mutable and Immutable Strings. Creating Strings. Operations on Strings. Threads: Creating Threads in Java. Java Thread Lifecycle. Multithreading in Java: Synchronization and Inter-process communication (IPC) in Threads. Applet: Java Applet class Architecture. Working and Lifecycle of Java Applet. Displaying text and animation, and passing parameters to Applet. Embedding Applets in a web page.	4

4.	Event-Driven Programming: Introducing the AWT: Window Fundamentals, Working with Frame Windows. Using AWT Controls: Control Fundamentals, Labels, Buttons, Applying Check Boxes, Lists, TextField. I/O Streams: Byte, Character, Buffered, Data, and Object Streams. Standard Streams. File I/O Basics, Reading and Writing to Files. Serializing Objects. Networking Classes and Interfaces: TCP/IP Server Sockets in Java, InetAddress, Whois. URL.	3
----	---	---

Text Books

1.	E. Balagurusamy, Programming with Java: A Primer, 4 th Edition, Tata Mcgraw Hill
2.	C.S. Horstmann and G. Cornell, Java 2 Vol-1 Fundamentals, Pearson Education

References

1.	Schildt, Java: The Complete Reference, 9th Edition, Tata McGraw Hill
2.	H.M. Dietel and P.J. Dietel, Java: How to Program, Pearson Education.
3.	K. Sierra, Sun Certified Programmer For Java 5, Wiley India.

Central University of Kashmir

Course Title	OOP Lab	Course Code	BTCS 410 L	
Degree	Bachelor of Technology	Branch	Computer Science & Engg.	
Course Type	Theory	Department	Information Technology	
Credits	02	L	T	P
Semester	4th	0	0	2

Course Objectives

- Students will be able to program using Java programming language.
- To use the Java in a variety of technologies and on different platforms.
- To develop software in the Java programming language

Learning Outcomes

- Knowledge of the structure and model of the Java programming language.
- Knowledge of object-oriented paradigm in the Java programming language.
- Design, analyse and evaluate computer programs using the Java programming language.
- Able to implement exception and basic event handling mechanisms in a program.
- Able to implement object-oriented concepts-constructors, encapsulation, inheritance and polymorphism

Course Synopsis

- Demonstrate the use of loops
- Implement programs on strings
- Implementation of all math class functions.
- Implementation of file management concept.

- Implementation of method over-riding and over loading.
- Implementation of inheritance.
- Multi-threaded programming
- Implement programs on exception handling
- Implementation of event handling mechanism

Course Outline / Content

Lab		Weeks
	• Program to calculate area and perimeter of the circle • Program on substring removal • Program for overloading functions in java • Program for hierarchical inheritance in java • Program for overriding • Program for abstract class • Program for interface. • Program for pre-defined exception handling • Program for user-defined exception handling • Program for Thread and exceptional handling • Program for thread synchronization • Program for Inter Thread Communication • Program for java utility (calendar class) • Program for string manipulation. • Programs on file streams • Program for applet with mouse listener	In acordance with theory of BTCS 404

	• Programs for frames and controls 14 • Programs for menus and dialog box • Programs for panel and layout • Programs for incorporating graphics	
--	--	--

Text Books

1.	E. Balagurusamy, Programming with Java: A Primer, 4 th Edition, Tata Mcgraw Hill.
2.	C.S. Horstmann and G. Cornell, Java 2 Vol-1 Fundamentals, Pearson Education.

References

1.	Schildt, Java: The Complete Reference, 9th Edition, Tata McGraw Hill
2.	H.M. Dietel and P.J. Dietel, Java: How to Program, Pearson Education
3.	K. Sierra, Sun Certified Programmer For Java 5, Wiley India.